Apna Campus (CollegeFlow)

System Architecture & Technology Stack Specification

Autonomous BPUT College ERP · Monorepo Architecture · 14 Role Portals

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1. Executive Architecture Overview

Apna Campus (CollegeFlow) is an enterprise-grade, autonomous college ERP and institutional management system designed specifically for autonomous engineering institutions (BPUT autonomous curriculum compliant). The solution is structured as a unified monorepo providing modular backend services, role-based security, and 14 dedicated interactive client portals.

Tier / Layer	Core Technology	Default Port	Architectural Role & Description
Presentation (Client)	React 18, Vite 6, Tailwind CSS	5173	SPA with 14 role-based portals, glassmorphism UI, client PDF generation, and real-time dashboard analytics.
Application (API)	NestJS 10, TypeScript 5.7	4000	Enterprise REST API gateway with dependency injection, Passport JWT, RBAC guards, global validation, and Swagger docs.
Auxiliary Feeds (API)	Express.js 4 (Node.js)	5000	Lightweight social feed microservice powering CampusConnect posts, media, and universal WhatsApp launcher.
Persistence (Data)	PostgreSQL 14+, Prisma ORM 5.22	5432	Relational database with strict foreign keys, enums, migrations, and idempotent seed generation.

2. Detailed Technology Stack Matrix

Layer	Technology / Library	Version	Purpose & Implementation Scope
Frontend	**React**	18.3.1	Component-based reactive UI architecture & virtual DOM
Frontend	**Vite**	6.0.5	High-performance ES module dev server & optimized production rollup builder
Frontend	**Tailwind CSS**	3.4.16	Utility-first modern styling framework & responsive design system
Frontend	**React Router DOM**	6.28.0	Client-side routing, navigation protection & portal view routing
Frontend	**Lucide React**	0.468.0	Modern, unified SVG iconography across all 14 portals
Frontend	**Recharts**	2.15.0	Interactive analytics for attendance, marks trends, and fee collection
Frontend	**jsPDF & AutoTable**	4.2 / 5.0	Client-side generation of fee receipts, hall tickets & grade sheets
Frontend	**SheetJS (XLSX)**	0.18.5	Excel spreadsheet parsing and roster/mark sheet export
Frontend	**Zod**	3.23.8	Type-safe schema validation for form submissions and inputs
Frontend	**Vitest & JSDOM**	2.1 / 26.0	Fast unit and browser-environment component testing
Backend	**NestJS Core / Common**	10.4.15	Modular DI framework, routing, controllers, services, and modules
Backend	**TypeScript**	5.7.2	End-to-end type safety, decorators, and interfaces across all modules

Layer	Technology / Library	Version	Purpose & Implementation Scope
Backend	**Passport & JWT**	0.7 / 4.0	Stateless Bearer JWT token strategy with claim verification
Backend	**bcryptjs**	2.4.3	Salted password hashing for credential security
Backend	**Swagger (OpenAPI)**	7.4.2	Automated OpenAPI 3.0 documentation generated at /api/docs
Backend	**class-validator & transformer**	0.14 / 0.5	Global ValidationPipe whitelist parsing, type casting & DTO contracts
Backend	**Multer**	2.4.0	Multipart file processing for payment proofs & student doc uploads
Backend	**RxJS**	7.8.1	Reactive stream operators for interceptors and async request handlers
Database	**PostgreSQL**	14+	ACID relational database engine with indexes, cascading deletes & enums
Database	**Prisma ORM**	5.22.0	Type-safe schema modeling, query engine & automated client generation
Database	**Prisma Migrate**	5.22.0	Committed version-controlled migration files in backend/prisma/migrations

3. Dedicated Role Portals & Access Scopes

#	Role Portal	Identifier / Access	Core Responsibilities & Capabilities
1	**Super Admin**	SYS-SUPER-01 / superadmin	Enterprise security, database backups, audit logs, system-wide role provisioning.
2	**Management**	GOV-CHAIR-01 / management	Executive financial dashboards, admission statistics, institutional KPI oversight.
3	**Principal**	OFF-PRIN-01 / principal	Autonomous institution governance, faculty performance, accreditation compliance.
4	**Admin**	ADM-001 / admin	Department master tables, staff onboarding, academic calendar and timetable scheduling.
5	**HOD**	HOD-CSE-001 / hod	Branch-specific performance metrics, faculty allocation, internal marks verification.
6	**Faculty**	FAC-042 / faculty	Attendance recording, continuous evaluation marks entry, missed class notes sharing.
7	**Student**	2305201001 / student	Academic profile, fee payment proofs, gate pass requests, semester results, timetable.
8	**Exam Cell**	EXM-001 / exam_cell	Autonomous examination scheduling, hall ticket generation, seating plans, grading rules.
9	**Accounts**	ACC-001 / accounts	Online fee verification, UPI/challan reconciliation, dues clearance, receipt issuance.
10	**Library**	LIB-001 / library	Book catalog management, RFID/barcode issue-return logs, digital journal access.
11	**Placement**	TPO-001 / placement	Campus recruitment drives, student eligibility filtering, interview rounds, offers.
12	**Transport**	TRN-001 / transport	Bus route planning, vehicle GPS fleet status, driver assignments, transport passes.
13	**Hostel / Warden**	HST-001 / hostel	Room allotment, digital gate pass approval, biometric in/out logs, mess feedback.
14	**Canteen / Staff**	CNT-001 / canteen	Cafeteria menu management, cashless student wallet transactions, order fulfillment.

4. Key Domain Subsystems & Business Logic

A. Role-Based Access Control (RBAC) & Security Architecture

The backend uses Passport JWT with a custom RolesGuard and @Roles() metadata decorator. Passwords are salted with bcryptjs. In the frontend, route protection and portal switching ensure users only view actions strictly within their privilege domain.

B. Digital Fee Payment Proof & Accounts Verification

Students upload payment receipts (UPI/NEFT/Challan) with transaction reference numbers. The Accounts section accesses a verification queue (/api/accounts/payment-verifications) to approve or reject with remarks, immediately updating ledger balances and triggering dynamic PDF fee receipts.

C. Examination Cell, Grading & Autonomous Marks Moderation

Supports internal assessment (30 marks) and end-term examinations (70 marks) tailored to BPUT autonomous engineering standards. Features automated SGPA/CGPA computation, grade conversion, backlog tracking, and PDF hall ticket generation with barcode verification.

D. Biometric Attendance & Dual-Step Hostel Gate Pass

Integrates real-time biometric tracking for hostel entry/exit. Gate pass requests initiate a multi-tier authorization workflow (Hostel Warden review -> Security Gate verification) complete with dynamic security QR codes and timestamped logging.

E. AI Campus Assistant (Apna AI Chatbot) & CampusConnect

Built-in intelligent campus virtual assistant answering student queries on syllabus, dues, exams, and policies. Augmented by CampusConnect social feed and universal WhatsApp launcher for automated student-faculty communication.

5. Infrastructure, DevOps & Cloud Readiness

Component	Target Platform / Tool	Configuration Details
Frontend Hosting	**Vercel / Static CDN**	Configured via vercel.json with SPA catch-all rewrites to /index.html.
Backend Hosting	**Render / Docker Node.js**	Configured via render.yaml; auto-detects cloud SSL mode (sslmode=require) for databases.
Database Hosting	**PostgreSQL (Render / Supabase / Neon)**	PostgreSQL connection string with automated pooling & SSL enforcement.
Monorepo Concurrency	**Concurrently 9.2**	Root npm run dev runs dev:backend and dev:frontend in parallel with color-coded logs.
Developer Onboarding	**PowerShell Script**	Automated setup via scripts/setup-dev.ps1 checking Node, npm, PostgreSQL, env & migrations.